

Journal of Biopharmaceutical Statistics

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/lbps20

A Bayesian group sequential design for randomized biosimilar clinical trials with adaptive information borrowing from historical data

Wen Zhang, Zhiying Pan & Ying Yuan

To cite this article: Wen Zhang, Zhiying Pan & Ying Yuan (2022) A Bayesian group sequential design for randomized biosimilar clinical trials with adaptive information borrowing from historical data, Journal of Biopharmaceutical Statistics, 32:3, 359-372, DOI: 10.1080/10543406.2022.2080700

To link to this article: https://doi.org/10.1080/10543406.2022.2080700

Published online: 09 Jun 2022.

Submit your article to this journal

Article views: 276

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=lbps20

JOURNAL OF BIOPHARMACEUTICAL STATISTICS
2022, VOL. 32, NO. 3, 359–372
https://doi.org/10.1080/10543406.2022.2080700

Taylor & Francis
Taylor & Francis Group

Check for updates

A Bayesian group sequential design for randomized biosimilar clinical trials with adaptive information borrowing from historical data

Wen Zhang [a], Zhiying Pan [b], and Ying Yuan [c]

[a]Department of Biostatistics and Data Science, The University of Texas Health Science Center at Houston, Houston, Texas, USA; [b]Amgen Inc, Thousand Oaks, California, USA; [c]Department of Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, Texas, USA

ABSTRACT

At the time of developing a biosimilar, the reference product has been on market for years and thus ample data are available on its efficacy and characteristics. We develop a Bayesian adaptive design for randomized biosimilar clinical trials to leverage the rich historical data on the reference product. This design takes a group sequential approach. At each interim, we employ the elastic meta-analytic-predictive (EMAP) prior methodology to adaptively borrow information from the historical data of the reference product to make go/no-go decision based on Bayesian posterior probabilities. In addition, the randomization ratio between the test and reference arms is adaptively adjusted at the interim with the goal to balance the sample size of the two arms at the end of trials. Simulation study shows that the proposed Bayesian adaptive design can substantially reduce the sample size of the reference arm, while achieving comparable power as the traditional randomized clinical trials that ignore the historical data. We apply our design to a biosimilar trial for treating breast cancer patients.

ARTICLE HISTORY
Received 12 June 2021
Accepted 29 March 2022

KEYWORDS
Bayesian adaptive design; Adaptive borrowing; Randomized trials; Elastic prior

1. Introduction

Biosimilar products offer additional treatment options and potentially lower healthcare costs for patients with fatal diseases such as cancer. US Food and Drug Administration (2015) recommends a totality of evidence approach based on a stepwise process for the development of biosimilars. To gain regulatory approval, a biosimilar must be comparable to the reference biologic in terms of structure, function, animal toxicity, human pharmacokinetics (PK) and pharmacodynamics (PD), immunogenicity, clinical safety, and efficacy (US Department of Health and Human Services 2014). It is essential to demonstrate that the biosimilar product has similar efficacy as the reference product through randomized clinical trials.

One important consideration when designing 2-arm randomized biosimilar trials is that by the time of developing a biosimilar, the reference biologic product has been usually approved by regulatory agencies and on market for years. It is of great interest to leverage these rich historical data to enhance the efficiency of the trial. Pan et al. (2017) proposed a calibrated power prior approach to adaptively borrow information from historical study for biosimilar trials. Weiss et al. (2018) proposed to use a meta-analytic-predictive (MAP) prior for the reference arm to borrow information from historical data, and an informative prior for the test arm based on analytical and clinical PK/PD data of the test product. Chen et al. (2011) proposed a Bayesian design to use a hierarchical modeling approach and power prior model to incorporate historical data for non-inferiority trials. Zhang et al.

CONTACT Ying Yuan yyuan@mdanderson.org Department of Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, Texas 77030, USA

© 2022 Taylor & Francis Group, LLC

(2021) proposed the elastic MAP (EMAP) prior to allow adaptive information borrowing from multiple historical datasets of the reference product based on the congruence of the observed trial data and historical datasets. Other related work, but not specifically for biosimilar trials, includes the power prior (Ibrahim et al. 2015), probability weighted power prior (Bennett et al. 2021), robust MAP prior (Schmidli et al. 2014), commensurate prior (Hobbs et al. 2011), and elastic prior (Jiang et al. 2021).

In this article, we propose a Bayesian group sequential design to adaptively borrow information from historical reference data for randomized biosimilar trials. Patients initially are 1:1 randomized into the test arm and the reference arm. At each interim, we employ the EMAP prior methodology to adaptively borrow information from the historical data of the reference product, and make go/no-go decision based on the posterior probability of the biosimilarity between the test and reference products. If the decision is "go," we determine the amount of information that is borrowed from the historical data, measured by the effective sample size (ESS), and adaptively adjust the subsequent randomization ratio to re-balance the sample size of the two arms. As a result, when the historical data are congruent to the trial data and thus more information can be borrowed, fewer patients will be randomized to the reference arm to save the sample size. The proposed method makes several new contributions. Compared to Pan et al. (2017), which focuses on borrowing information from a single historical dataset, our method is capable of leveraging information from multiple historical datasets. In addition, the design of Pan et al. (2017) assumes a fixed sample size. In contract, our design adaptively adjusts the randomization ratio and re-balances the sample size of the two arms based on the amount of information borrowed. Compared to Weiss et al. (2018) and Zhang et al. (2021), which focus on the data analysis of biosimilar trials, and Chen et al. (2011), which focuses on sample size determination, our design provides a group sequential approach that allows the interim decision of early stopping unpromising biosimilars, and a mechanism to adaptively adjust the sample size and information borrowing from historical data.

This article proceeds as follows. Section 2 introduces the EMAP prior and describes the adaptive trial design. Section 3 applies the proposed design to a biosimilar trial and compares it with the conventional design. Sensitivity analyses are also conducted to assess the robustness of the proposed design. Section 4 closes this article with a brief discussion.

2. Method

2.1. Elastic meta-analytic-predictive (EMAP) prior

Let y denote a binary efficacy endpoint (e.g., response/no response) with $y = 1$ indicating the favorable outcome (e.g., response), and $\mu_r = \mathrm{E}(y)$ denote the response rate for the reference arm. Let $D = (x_r, n_r)$ denote the observed interim data from the reference arm, where $x_r = \sum_{i=1}^{n_r} y_i$ and n_r denote the number of responses and sample size, respectively. Let $D_{\mathbf{0}} = (D_{01}, \cdots, D_{0J})$ denote J independent historical datasets for the reference product with $D_{0j} = (x_j, n_j)$, where n_j is the sample size and x_j is the number of responses.

The EMAP prior (Zhang et al. 2021) is an extension of MAP prior (Schmidli et al. 2014; Weiss et al. 2018). We first introduce the latter. The MAP prior incorporates the prior information from $D_{\mathbf{0}}$, based on the following meta-analysis model:

$$x_j|\mu_j \sim Binomial(n_j, \mu_j) \tag{1}$$

$$\mu_j|\alpha_0, \beta_0 \sim Beta(\alpha_0, \beta_0), \tag{2}$$

where μ_j is the response rate for D_{0j}, $j = 1, \cdots, J$, and $\alpha_0, \beta_0 > 0$ are unknown parameters. To facilitate the specification of priors for α_0 and β_0, we reparameterize them using the mean (μ_0) and variance (τ_0^2) parameters (Weiss et al. 2018),

$$\mu_0 = \frac{\alpha_0}{\alpha_0 + \beta_0} \tag{3}$$

$$\tau_0^2 = \frac{\mu_0(1-\mu_0)}{\alpha_0 + \beta_0 + 1}, \tag{4}$$

As both μ_0 and τ_0 are bounded between 0 and 1, we assign them beta priors $\mu_0 \sim Beta(\alpha_\mu, \beta_\mu)$ and $\tau_0 \sim Beta(\alpha_\tau, \beta_\tau)$, where $\alpha_\mu, \beta_\mu, \alpha_\tau$ and β_τ are hyperparameters.

By treating the current reference arm as the $(J+1)$th study of the meta-analysis, the MAP prior is given by $p(\mu_r|D_0) \equiv p(\mu_{(J+1)}|D_0)$. This posterior predictive distribution does not have a closed form, but can be approximated using a beta distribution (Weiss et al. 2018)

$$\mu_r|\mathbf{D}_0 \sim Beta(\alpha, \beta), \tag{5}$$

where α and β are determined by matching the first two moments of $Beta(\alpha, \beta)$ with those of $p(\mu_{(J+1)}|D_0)$. When necessary, a mixture of beta distributions can be used to improve the accuracy of approximation (Schmidli et al. 2014).

One limitation of the MAP prior is that it always executes full information borrowing from D_0, which may cause substantial type I error inflation and bias when D and D_0 are incongruent. To address this issue, Zhang et al. (2021) proposed the EMAP prior to achieve adaptive information borrowing according to the congruence between D and D_0. The idea of the EMAP is straightforward: discount the MAP prior adaptively according to an elastic function whose value depends on the congruence between D and D_0. Specifically, let $S > 0$ denote a statistic that measures the congruence between D and D_0. A reasonable choice is the scaled binomial test statistic:

$$S = n_r^{-\frac{1}{4}} \frac{|\bar{y}_r - \mu_0|}{\sqrt{\frac{\mu_0(1-\mu_0)}{n_r}}}, \tag{6}$$

where $n_r^{-\frac{1}{4}}$ is a scale factor added for the property of information-borrowing consistency, which means that when the sample size of the reference arm of the current trial $n_r \to \infty$, it achieves full information borrowing if D_0 and D are congruent (i.e., $\mu_0 = \mu_r$) and does not borrow any information if D_0 and D are incongruent (i.e., $\mu_0 \neq \mu_r$). As a result, the EMAP prior has desirable property that its asymptotically maintains the type I error at the nominal value, see Zhang et al. (2021) for details. The choice of S is not unique, other statistics (e.g., Chi-square statistic, Bayesian predictive p-value (Gelman et al. 1995)) could also be used. Zhang et al. (2021) showed that the choice of S is not critical, as long as it appropriately captures the heterogeneity between D_0 and D.

The EMAP prior is obtained by inflating the variance of the full-information MAP prior (5) by a factor of $g(S)$, defined as

$$\mu_r|D_0 \sim Beta(\alpha g(S), \beta g(S)),$$

where $g(S)$ is the elastic function, taking a form of logistic function as follows:

$$g(S) = \frac{1}{1 + \exp\{a + b \times \log(S)\}}, \tag{7}$$

where a and $b > 0$ are tuning parameters. Compared to the MAP prior (5), the key of the EMAP prior is that the information contained in the prior is proactively moderated by the elastic function according to the congruence between D and D_0 (see Figure 1). When D and D_0 are congruent, $S \to 0$ (with a large sample) and the EMAP prior becomes the MAP, achieving full information borrowing; and when D and D_0 are not congruent, $S \to \infty$ (with a large sample) and the EMAP prior becomes non-informative with ESS = 0, refraining from any information borrowing. As the EMAP prior depends on the observed trial data D (through S), thus it may be regarded as an empirical Bayes method.

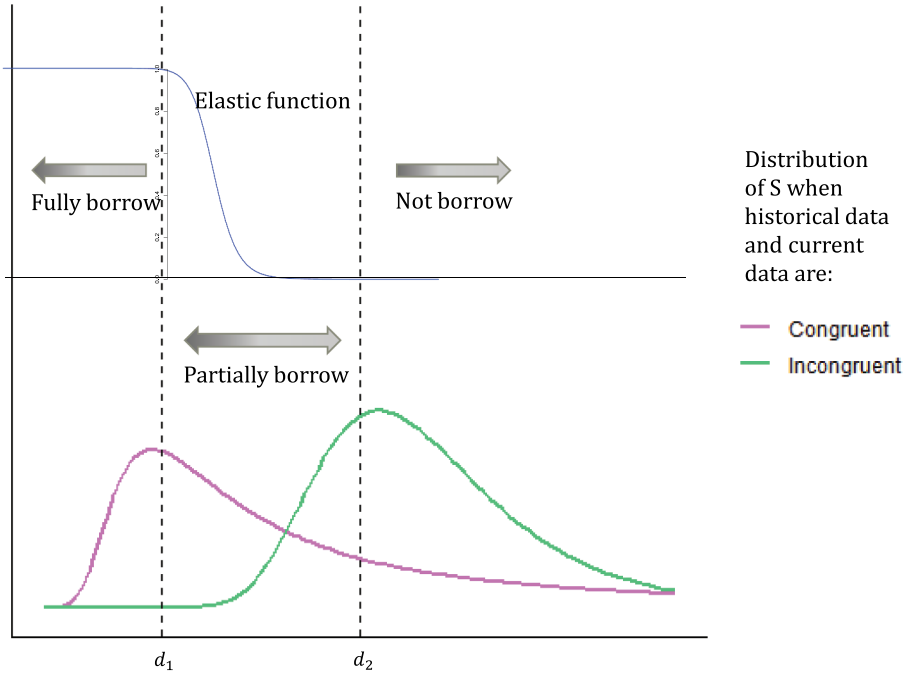


Figure 1. The EMAP prior achieves adaptive information borrowing through the elastic function. d_1 and d_2 are two thresholds where the elastic function enforces full information borrowing and no information borrowing.

The choice of the elastic function is not unique, other reasonable monotonic function can also be used. A special and limiting form of the logistic elastic function (7), obtained with a large value of b, is the following step function

$$g(S) = \begin{cases} 1 & S \leq d \\ 0 & S > d \end{cases} \tag{8}$$

where d is a cutoff, which can be calibrated using the procedure described in the Appendix. This elastic function executes full information borrowing if $S \leq d$ and doesn't borrow information if $S > d$. Numerical studies show that this step function could achieve comparable operating characteristics as the smooth elastic function in formula (7). The advantage of the step elastic function is that it is simple to calibrate because of the use of a single cutoff parameter. Zhang et al. (2021) and Jiang et al. (2021) recommended the use of step elastic function. Reimherr et al. (2021) provided the theory support of using the step function by showing that the gain of borrowing information (or the effective sample size of the prior) becomes negative at a certain point in the presence of the prior-likelihood conflict. The procedure to calibrate the step elastic function (i.e., the value of d) is provided in the Appendix. Because its proactive and adaptive control of information borrowing according to the congruence between D and D_0, EMAP outperforms the robust MAP (Schmidli et al. 2014) with lower type I error and comparative power (Zhang et al. 2021).

2.2. EMAP trial design

We propose an adaptive group sequential, two-arm randomized biosimilar trial design based on the EMAP method (AEMAP design). The trial starts with 1:1 randomization to the test arm and the reference arm. At each interim, we apply the EMAP method to adaptively borrow information from

historical reference data D_0 to make go/no-go decision. The trial will be early terminated (i.e., no-go) if there is strong evidence against the biosimilarity between the the test arm and the reference arm. If the interim decision is "go," depending on the amount of information we can borrow from D_0, measured by the ESS, we prospectively adjust the randomization ratio for the next cohort with the goal to balance the sample size between two arms at next interim analysis and the end of the trial.

2.2.1. Bayesian go/no-go decision rule

We first describe our decision rule for interim go/no-go decisions. Let μ_r and μ_t denote the response rates for the reference and test arms, respectively. The objective of the biosimilar trial is to test whether the test product is equivalent to the reference product.

$$H_0 : \mu_t - \mu_r \le \lambda_L \quad or \quad \mu_t - \mu_r \ge \lambda_U, \quad versus \quad H_1 : \lambda_L < \mu_t - \mu_r < \lambda_U, \tag{9}$$

where λ_L and λ_U are the prespecified biosimilar margins. At each interim analysis, the go/no-go decision is made based on the following Bayesian criterion:

$$\text{Stop the trial if } \Pr(\lambda_L < \mu_t - \mu_r < \lambda_U \,|\, data) < C_f; \text{ otherwise, continue.} \tag{10}$$

C_f is a probability cutoff for futility stopping, calibrated through simulations to ensure a desirable operating characteristics (e.g., a certain percentage of early termination under H_0). If the trial is not early terminated, then at the end of trial, we reject H_0 and conclude that test product is equivalent to reference product using the following criterion:

$$\Pr(\lambda_L < \mu_t - \mu_r < \lambda_U \,|\, data) > C_s,$$

where C_s is a probability cutoff, calibrated through simulations to ensure desirable operating characteristics (e.g., type I error).

At each interim, we apply the EMAP prior method to the reference arm to borrow information from D_0, resulting in $p(\mu_r|D, D_0)$, the posterior distribution of μ_r as

$$Beta(\alpha g(S) + x_r, \beta g(S) + n_r - x_r).$$

Unlike the reference arm, typically there is limited prior information on the test arm, thus we apply the the noninformative prior $\mu_t \sim Beta(\alpha_t, \beta_t)$ to the test arm, where α_t and β_t are small values such as $\alpha_t = \beta_t = 0.5$. Given the observed interim data $D_t = (x_t, n_t)$ in the test arm, the posterior distribution of μ_t, $p(\mu_t|D_t)$ is given by

$$Beta(\alpha_t + x_t, \beta_t + n_t - x_t).$$

As the test arm and the reference arm are independent, the posterior probability used in the Bayesian decision-making criterion is calculated by

$$\Pr(\lambda_L < \mu_t - \mu_r < \lambda_U \,|\, D_t, D, D_0) = \iint I(\lambda_L < \mu_t - \mu_r < \lambda_U)\, p(\mu_r \,|\, D, D_0)\, p(\mu_t \,|\, D_t) d\mu_r d\mu_t.$$

2.2.2. Trial design with sample size re-balancing

When the interim decision is "go," according to the futility stopping rule (10), we adaptively adjust the randomization ratio between the test arm and the reference arm to re-balance the sample size of the two arms, according to the amount of information that the reference arm borrows from D_0. The rationale is that if substantial information can be borrowed from D_0, we should randomize fewer patients to the reference arm to save sample size. This sample size re-balancing process will be performed at each interim in light of accrued data. The goal is to achieve the same ESS in two arms at the end of the trial. For the test arm, as no information is borrowed, the ESS is the actual sample size. For the reference arm, the ESS is the actual sample size plus the ESS of the EMAP prior. One advantage

of the EMAP prior is that it is simple to quantify the information of the prior using the ESS. It is well known that under the beta-binomial model with the prior $Beta(\alpha, \beta)$, $\alpha + \beta$ can be interpreted as the ESS. Thus, the ESS of the EMAP prior is simply given by $g(S)(\alpha + \beta)$.

Suppose that the target ESS for each arm is N, which is chosen based on the standard sample size calculation to ensure appropriate power for evaluating the biosimilarity. The trial has a total of (K-1) interims, dividing patient accrual into K cohorts. For notational brevity, we assume that the interims are equally spaced, but our method directly applies to non-equally spaced interims. That is, each cohort has $\eta = N/K$ patients in the reference arm, and we perform the k th interim analysis when the k- th cohort are accrued, $k = 1, \ldots, K-1$, and the final analysis when all K cohorts are accrued. To mimic the conventional 1:1 randomization (without using historical data), we set the target ESS for each cohort to be η per arm. Assume at the k th interim, let S_k denote the value of S evaluated based on the interim data. Because the EMAP prior contributes an ESS of $g(S_k)(\alpha + \beta)$, the target accrual for the $(k+1)$ th cohort of the reference arm is

$$m_{k+1} = max(\eta - g(S_k)(\alpha + \beta), 0). \tag{11}$$

In the case that D and D_0 are highly incongruent, $g(S_k) \to 0$ and thus little information will be borrowed from D_0. The randomization will proceed as the standard RCT with $m_{k+1} = \eta$. In summary, the proposed AEMAP design is comprised of the following steps:

1. Enroll 2η patients and randomize them 1:1 to the test and reference arms. Typically, $\eta = 1/3N$ to $1/2N$ is a reasonable choice, which allows reasonable amount of data for the first interim analysis.

3. At kth $(k = 1, \cdots, K-1)$ interim analysis, compute the posterior probability $p = \Pr(\lambda_L < \mu_t - \mu_r < \lambda_U | D_t, D, D_0)$.

(a) If $p < C_f$, terminate the trial early and conclude that test product and reference product are not similar.

(b) Otherwise, enroll additional $(\eta + m_{k+1})$ patients (i.e., $(k+1)$th: cohort) and randomize them to the test arm and the reference arm in the ratio of $\eta : m_{k+1}$, where m_{k+1} is given by (11).

3. Stop the trial when the ESS of the two arms reaches N, and compute the posterior probability $\Pr(\lambda_L < \mu_t - \mu_r < \lambda_U | D_t, D, D_0)$. If $\Pr(\lambda_L < \mu_t - \mu_r < \lambda_U | D_t, D, D_0) > C_s$, we conclude that test product and the reference product are biosimilar.

In Step 2(a), we consider the futility stopping. When appropriate, the early stopping rule for biosimilarity can be added: If $p > C_s$, terminate the trial early and conclude that test product and reference product are similar. The cutoff C_s should be calibrated using simulation to ensure desirable type I error rate.

4. Simulation studies

4.1. Operating characteristics

We apply the AEMAP design to a two-arm randomized biosimilar trial with trastuzumab as the reference product. Trastuzumab is a monoclonal antibody for treating HER2 receptor positive patients with breast or stomach cancer. Six historical datasets of trastuzumab are available, as shown in Table 1. The primary endpoint is a binary variable, indicating whether a patient has pathologic complete response (pCR) after treatment. The pCR rate in the historical datasets varies from 0.38 to 0.58 with the sample size ranging from 42 to 307.

We first performed a meta-analysis to synthesize the historical data based on model (1)-(4). We assign priors to μ_0 and τ_0 to reflect the plausible range of the pCR rate. Specifically, we set the prior mean and variance of μ_0 to be $\mathrm{E}(\mu_0) = 0.5$ and $\mathrm{Var}(\mu_0) = (0.2 * 0.5)^2$, representing a prior point estimate of 0.5, and expectation that μ_0 is within 0.2 of mean with probability of roughly 0.95. We give τ_0 a prior point estimate and a prior variance of $\mathrm{E}(\tau_0) = 0.1$ and $\mathrm{Var}(\tau_0) = 0.1^2$. We solve the above

Table 1. Historical data from six clinical trials of trastuzumab with pathologic complete response (pCR) as the primary endpoint.

Study(Author and year)	Number of pCR	Sample size
ABCSG-24 (Steger et al. 2009)	17	42
CALGB (Carey et al. 2013)	48	120
NOAH (Gianni et al. 2010)	45	117
GeparQuinto (Untch et al. 2012)	137	307
HannaH (Ismael et al. 2012)	121	298
Bayraktar et al. (2012)	137	235

equations and get priors $\mu_0 \sim Beta(12, 12)$ and $\tau_0 \sim Beta(0.8, 7.2)$. Given the priors and fit the meta-analysis model, the resulting MAP prior is $\mu_r \sim Beta(122.02, 173.59)$ with the posterior mean $\hat{\mu}_0 = 0.413$.

In this trial, the biosimilar margin is 0.1, i.e., the test product is considered to be biosimilar to trastuzumab if $-0.1 < \mu_t - \mu_r < 0.1$. Applying standard sample size calculation method, $N = 510$ per arm is required to obtain 90% power at the significance level of 0.05, assuming no information is borrowed from historical data and $\mu_t = \mu_r = \hat{\mu}_0 = 0.413$. Two interims are planned when the sample size of the test arm reaches $1/3N$ and $2/3N$. The step elastic function in formula (8) is used to construct the EMAP prior. We evaluate the operating characteristics of the AEMAP design under various scenarios where D are congruent or incongruent to D_0. For example, Scenarios 1–3 represent the congruent cases with true response rates of reference arm $\mu_r = 0.413$ (i.e., $\mu_r = \hat{\mu}_0$), 0.392 and 0.434 (i.e., $\mu_r = \hat{\mu}_0 \pm 0.021$); and Scenarios 8–9 represent the incongruent cases with $\mu_r = 0.313$ and 0.513 (i.e., $\mu_r = \hat{\mu}_0 \pm 0.1$). We considered several response rates for the test arm: $\mu_t = \mu_r$ (i.e., the test and reference products are biosimilar), $\mu_r \pm 0.1$ and $\mu_r \pm 0.15$ (i.e., the test and reference products are not biosimilar). Under each scenario, 1000 trials are simulated. We compared the AEMAP to group sequential design (GS) without borrowing information from D_0, and the standard randomized controlled trial (RCT) design with no interim analyses.

Table 2 summarizes the results of simulation. In Scenarios 1–4, $\mu_t - \mu_r$ is larger than or equal to the margin 0.1, so the percentage of claiming biosimilarity (PCB) represents type I error. In Scenarios 1 and 2, with $\mu_t - \mu_r = \pm 0.1$, AEMAP, GS and RCT all control type I error rates $< 5\%$. AEMAP and GS both have percentages of early termination (PET) at around 11%. AEMAP offers substantial sample

Table 2. Simulation results of the AEMAP design with 2 interim futility analyses, compared to the randomized controlled trial (RCT) design (without futility stopping), and group sequential (GS) design with 2 interim futility analyses. PCB denotes the percentage of claiming biosimilarity (i.e., rejecting the null hypothesis), and PET denotes the percentage of early termination.

			PCB			PET		Sample size		
Scenario	μ_r	μ_t	RCT	GS	AEMAP	GS	AEMAP	RCT	GS	AEMAP
Congruence										
1	0.413	$\mu_r - 0.1^*$	0.053	0.049	0.033	0.118	0.111	1020	955	715
2	0.413	$\mu_r + 0.1^*$	0.046	0.053	0.033	0.114	0.113	1020	957	716
3	0.413	$\mu_r - 0.15^*$	0.000	0.001	0.000	0.537	0.622	1020	729	535
4	0.413	$\mu_r + 0.15^*$	0.001	0.000	0.001	0.505	0.574	1020	744	557
5	0.413	μ_r	0.888	0.897	0.935	0.000	0.001	1020	1020	762
6	0.392	μ_r	0.896	0.897	0.906	0.004	0.001	1020	1020	774
7	0.434	μ_r	0.890	0.879	0.892	0.000	0.001	1020	1020	784
Incongruence										
8	0.513	$\mu_r - 0.1^*$	0.054	0.054	0.105	0.114	0.123	1020	956	921
9	0.313	$\mu_r + 0.1^*$	0.049	0.051	0.100	0.110	0.119	1020	959	926
10	0.513	$\mu_r - 0.15^*$	0.001	0.001	0.016	0.497	0.510	1020	755	708
11	0.313	$\mu_r + 0.15^*$	0.000	0.000	0.015	0.511	0.520	1020	743	708
12	0.513	μ_r	0.881	0.870	0.840	0.000	0.004	1020	1020	990
13	0.313	μ_r	0.924	0.935	0.883	0.000	0.003	1020	1020	991

*PCB is type I error

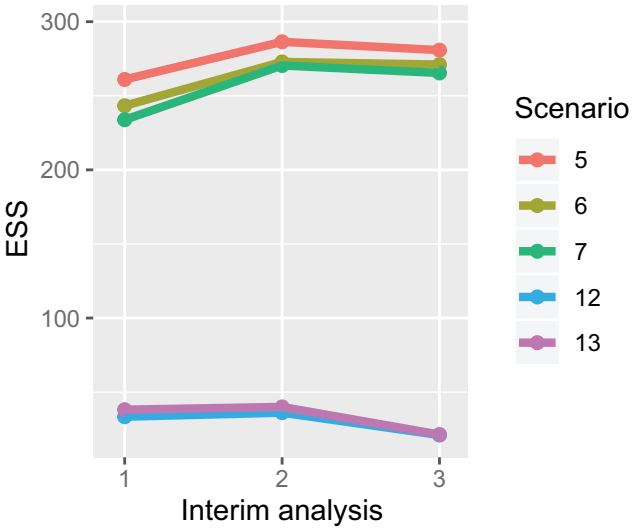


Figure 2. Effective sample size (ESS) of the EMAP prior at each interim analysis, under simulation Scenarios 5, 6, 7, 12 and 13.

size saving. Compared to GS, AEMAP cuts the total sample size from 954.79 to 714.50 in Scenario 1. And from 956.83 to 715.86 in Scenario 2. In Scenarios 3–4, $\mu_t - \mu_r = \pm 0.15$, which is larger than margin 0.1. Both GS and AEMAP designs early terminate the trial with high percentages (53.7% and 50.5% in GS; 62.2% and 57.4% in AEMAP), resulting in substantially smaller sample sizes than RCT. Compared to GS, AEMAP offers about 27% saving in sample size. Because $|\mu_t - \mu_r|$ is greater than the margin used to design the study, the type I errors of all designs are lower than the nominal value and close to 0. In Scenarios 5–7, $\mu_t = \mu_r$ and thus the PCB represents power. AEMAP yields comparable or higher power than GS and RCT, and saves about 258 patients (762 vs. 1020) in Scenario 5, about 247 patients (about 773 vs. 1020) in Scenarios 6, and about 237 patients (about 783 vs. 1020) in Scenarios 7.

Scenarios 8–13 represent the cases that D are incongruent to D_0. In Scenarios 8–11, $\mu_t - \mu_r$ is larger than or equal to the margin 0.1, and the PCB represents type I error. In Scenarios 8–9, $\mu_t - \mu_r = 0.1$, and both AEMAP and GS designs have a PET in the range of 11%-12%. The sample size of AEMAP is about 4% less than that of GS. The sample size difference between this two designs are much smaller than that in the congruent scenarios (i.e., Scenarios 1–7). This shows that AEMAP is capable to sense the incongruence between D and D_0 and adaptively adjust the strength of information borrowing. Under the finite sample, due to randomness, D and D_0 may be very similar although their underlying true treatment effects are different. Thus, AEMAP still induces some, although much weaker, information borrowing in Scenarios 8–9. As a result, AEMAP has higher type I error than GS. Similar results are observed in Scenarios 10–11. In Scenarios 12–13, $\mu_t = \mu_r$ and the PCB represents power of the designs. The power of AEMAP is slightly lower than GS and RCT, but the sample size is also slightly smaller.

Figure 2 shows the average ESS of historical data borrowed by AEMAP at each interim. In general, ESS is stable over interims. It is large (>230) when D and D_0 are congruent (e.g., Scenarios 5, 6 and 7); and small (<40) when D and D_0 are not congruent (e.g., Scenarios 12 and 13). This result shows the adaptation of the AEMAP and is consistent with the result that AEMAP provides substantial sample size saving when D and D_0 are congruent.

4.2. Sensitivity analyses

We conduct sensitivity analysis to assess the robustness of the AEMAP design with 1 mid-trial interim analysis at $N/2$. Table 3 shows the simulation results. The results are generally similar to these with 2 interims (Table 2). That is, AEMAP yields comparable power as the GS and RCT with substantial

Table 3. Simulation results of the AEMAP design with 1 interim futility analysis, compared to the randomized controlled trial (RCT) design (without futility stopping), and group sequential (GS) design with 1 interim futility analysis. PCB denotes the percentage of claiming biosimilarity (i.e., rejecting the null hypothesis), and PET denotes the percentage of early termination.

			PCB			PET		Sample size		
Scenario	μ_r	μ_t	RCT	GS	AEMAP	GS	AEMAP	RCT	GS	AEMAP
Congruence										
1	0.413	$\mu_r - 0.1^*$	0.053	0.050	0.043	0.101	0.103	1020	969	749
2	0.413	$\mu_r + 0.1^*$	0.046	0.055	0.037	0.108	0.112	1020	966	747
3	0.413	$\mu_r - 0.15^*$	0.000	0.0004	0.000	0.457	0.610	1020	787	617
4	0.413	$\mu_r + 0.15^*$	0.001	0.0004	0.000	0.454	0.574	1020	789	628
5	0.413	μ_r	0.888	0.909	0.919	0.000	0.000	1020	1020	781
6	0.392	μ_r	0.896	0.905	0.906	0.001	0.001	1020	1020	796
7	0.434	μ_r	0.890	0.897	0.899	0.001	0.000	1020	1020	800
Incongruence										
8	0.513	$\mu_r - 0.1^*$	0.054	0.051	0.094	0.106	0.119	1020	967	939
9	0.313	$\mu_r + 0.1^*$	0.049	0.051	0.093	0.096	0.123	1020	972	937
10	0.513	$\mu_r - 0.15^*$	0.001	0.000	0.013	0.486	0.514	1020	785	753
11	0.313	$\mu_r + 0.15^*$	0.000	0.000	0.011	0.492	0.492	1020	791	751
12	0.513	μ_r	0.881	0.878	0.853	0.000	0.003	1020	1020	1000
13	0.313	μ_r	0.924	0.933	0.894	0.000	0.002	1020	1020	1001

*PCB is type I error

sample size saving when D and D_0 are congruent. When the test product and reference product are not biosimilar (e.g., Scenarios 1–4 and 8–11), the sample size with 1 interim is larger than that with 2 interims. This is because the trial has higher probability to be terminated early in the latter. We also evaluate the performance of proposed design with different choice of the congruence measure S (e.g., using Bayesian predictive p value), and design parameters used in the calibration procedure of the EMAP prior method. The results (see Appendix) show that the AEMAP design is generally robust.

5. Discussion

We have proposed a group sequential design for biosimilar clinical trials that adaptively borrows information from historical data based on the EMAP prior method. The AEMAP design is comprised of several interim analyses. At each interim, the "go/no-go" decision is made based on a Bayesian stopping criterion, and the amount of information borrowed from historical data is re-evaluated to adjust the randomization ratio and required sample size for the reference arm. Simulation studies show that compared to standard RCT design, the proposed AEMAP design offers significant sample size saving when the historical data are congruent to the trial data, with similar power.

To maximize the benefit of the AEMAP design, at the trial design phase, it is critically important to carefully select appropriate historical data D_0 such that there is high likelihood that D and D_0 are congruent, e.g., choose D_0 that have similar target population, same endpoint, and treatment procedures as the current trial. If there is a high likelihood that D and D_0 are not congruent, e.g., due to different patient enrollment eligibility criteria, the standard RCT design may be preferred, without borrowing information from D_0. The AEMAP design is able to adaptively adjust its information borrowing based on the congruence between D and D_0. However, under finite samples, it cannot completely prevent, although substantially weakens, the (inappropriate) information borrowing when D and D_0 are incongruent.

One extension to maximize the benefit of the AEMAP design is to leverage covariate information, if available, to increase the similarity between D and D_0. This can be done by first applying propensity score matching methods (Austin 2011; Rosenbaum and Rubin 1983) to select a subset of historical data, and then apply the AEMAP design. This paper focuses on the binary endpoint. AEMAP can be readily extended to accommodate the other types of endpoints. This is greatly facilitated by the Bayesian approach used by AEMAP. The major change required is to replace models (1) and (2)

with appropriate model, e.g., normal hierarchical model for continuous endpoints and piecewise exponential or other survival model for the time-to-event endpoints, with appropriate congruence statistics, the decision rule of the design largely remains the same.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

ORCID

Wen Zhang http://orcid.org/0000-0003-2175-2888

References

Austin, P. C. 2011. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behavioral Research* 46 (3):399–424. doi:10.1080/00273171.2011.568786.

Bayraktar, S., A. M. Gonzalez-Angulo, X. Lei, A. U. Buzdar, V. Valero, A. Melhem-Bertrandt, H. M. Kuerer, G. N. Hortobagyi, A. A. Sahin, and F. Meric-Bernstam. 2012. Efficacy of neoadjuvant therapy with trastuzumab concurrent with anthracycline-and nonanthracycline-based regimens for HER2-positive breast cancer. *Cancer* 118 (9):2385–2393.

Bennett, M., S. White, N. Best, and A. Mander. 2021. A novel equivalence probability weighted power prior for using historical control data in an adaptive clinical trial design: A comparison to standard methods. *Pharmaceutical Statistics* 20 (3):462–484. doi:10.1002/pst.2088.

Carey, L. A., D. A. Berry, D. Ollila, L. Harris, I. E. Krop, D. Weckstein, N. L. Henry, C. K. Anders, C. Cirrincione, E. P. Winer, et al. 2013. Clinical and translational results of CALGB 40601: A neoadjuvant phase III trial of weekly paclitaxel and trastuzumab with or without lapatinib for HER2-positive breast cancer. *Journal of Clinical Oncology* 31 (suppl):500a.

Chen, M. H., J. G. Ibrahim, P. Lam, A. Yu, and Y. Zhang. 2011. Bayesian design of noninferiority trials for medical devices using historical data. *Biometrics* 67 (3):1163–1170. doi:10.1111/j.1541-0420.2011.01561.x.

Gelman, A., J. B. Carlin, H. S. Stern, and D. B. Rubin. 1995. Bayesian data analysis Chapman and Hall. *CRC Texts in Statistical Science*.

Gianni, L., W. Eiermann, V. Semiglazov, A. Manikhas, A. Lluch, S. Tjulandin, M. Zambetti, F. Vazquez, M. Byakhow, M. Lichinitser, et al. 2010. Neoadjuvant chemotherapy with trastuzumab followed by adjuvant trastuzumab versus neoadjuvant chemotherapy alone, in patients with HER2-positive locally advanced breast cancer (the NOAH trial): A randomised controlled superiority trial with a parallel HER2-negative cohort. *The Lancet* 375 (9712):377–384.

Hobbs, B. P., B. P. Carlin, S. J. Mandrekar, and D. J. Sargent. 2011. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics* 67 (3):1047–1056. doi:10.1111/j.1541-0420.2011.01564.x.

Ibrahim, J. G., M. H. Chen, Y. Gwon, and F. Chen. 2015 December 10. The power prior: Theory and applications. *Statistics in Medicine* 34(28): 3724–3749. doi:10.1002/sim.6728. Epub 2015 Sep 7. PMID: 26346180; PMCID: PMC4626399.

Ismael, G., R. Hegg, S. Muehlbauer, D. Heinzmann, B. Lum, S. B. Kim, T. Pienkowski, M. Lichinitser, V. Semiglazov, B. Melichar, et al. 2012. Subcutaneous versus intravenous administration of (neo) adjuvant trastuzumab in patients with HER2-positive, clinical stage I–III breast cancer (HannaH study): A phase 3, open-label, multicentre, randomised trial. *The Lancet Oncology* 13 (9):869–878. doi:10.1016/S1470-2045(12)70329-7.

Jiang, L., L. Nie, and Y. Yuan. 2021. Elastic priors to dynamically borrow information from historical data in clinical trials. *Biometrics*.

Pan, H., Y. Yuan, and J. Xia. 2017. A Calibrated Power Prior Approach to borrow information from historical data with application to biosimilar clinical trials. *Applied Statistics* 66:979–996.

Reimherr, M., X. L. Meng, and D. L. Nicolae. 2021. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society: Series B* 83 (3):413–437. doi:10.1111/rssb.12414.

Rosenbaum, P. R., and D. B. Rubin. 1983. The central role of the propensity score in observational studies for causal effects. *Biometrika* 70 (1):41–55. doi:10.1093/biomet/70.1.41.

Schmidli, H., S. Gsteiger, S. Roychoudhury, A. O'Hagan, D. Spiegelhalter, and B. Neuenschwander. 2014. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics* 70 (4):1023–1032. doi:10.1111/biom.12242.

Steger, G., R. Greil, R. Jakesz, A. Lang, B. Mlineritsch, E. Melbinger-Zeinitzer, C. Marth, H. Samonigg, E. Kubista, and M. Gnant. 2009. Final results of ABCSG-24, a Randomized Phase III Study Comparing Epirubicin, Docetaxel, and Capecitabine (EDC) to Epirubicin and Docetaxel (ED) as Neoadjuvant Treatment for Early Breast Cancer and Comparing ED/EDC+ Trastuzumab (T) to ED/EDC as Neoadjuvant Treatment for Early HER-2 Positive Breast Cancer. *Cancer Research*, 69 (24 Suppl.Abstract Nr):1081.

Untch, M., S. Loibl, J. Bischoff, H. Eidtmann, M. Kaufmann, J. U. Blohmer, J. Hilfrich, D. Strumberg, P. A. Fasching, R. Kreienberg, et al. 2012. Lapatinib versus trastuzumab in combination with neoadjuvant anthracycline-taxane-based chemotherapy (GeparQuinto, GBG 44): A randomised phase 3 trial. *The Lancet Oncology* 13 (2):135–144. doi:10.1016/S1470-2045(11)70397-7.

US Department of Health and Human Services. 2014. Clinical pharmacology data to support a demonstration of biosimilarity to a reference product. *Center for Drug Evaluation and Research (CDER) and Center for Biologics Evaluation and Research (CBER)*. Rockville, MD

US Food and Drug Administration. 2015. Guidance for industry: Scientific considerations in demonstrating biosimilarity to a reference product. *Center for Drug Evaluation and Research (CDER) and Center for Biologics Evaluation and Research (CBER)*. Rockville, MD

Weiss, R. E., X. Xia, N. Zhang, H. Wang, and E. Chi. 2018. Bayesian methods for analysis of biosimilar phase III trials. *Statistics in Medicine* 37 (20):2938–2953. doi:10.1002/sim.7814.

Yin, G, and Y, Yuan. 2009. Bayesian dose finding in oncology for drug combinations by copula regression. *Journal of the Royal Statistical Society Series C (Applied Statistics)*, 58(2): 211–224.

Zhang, W., Z. Pan, and Y. Yuan. 2021. Elastic meta-analytic-predictive prior for dynamically borrowing information from historical data with application to biosimilar clinical trials. *Contemporary Clinical Trials* 110:106559. doi:10.1016/j.cct.2021.106559.

Appendix

A.1. Calibration procedure in step elastic function to find cutoff d and utility search process for percentile ρ_1

A special and limiting form of the logistic elastic function (7), obtained with a large value of b, is the step function in (8). The advantage of the step elastic function is that it is simple to calibrate because of the use of a single cutoff parameter d. The calibration procedure of parameter d is as follows: Generate 10,000 replicates of the reference arm data $D = (x, n)$ from $x \sim Binomial(n, \hat{\mu}_0)$, and calculate statistic S for each replicate, where $\hat{\mu}_0$ is the posterior mean of historical data based on meta-analysis model. d can be found as the ρ_1 percentile of the empirical distribution of S.

A percentile ρ_1 is needed to define the borrowing and non-borrowing regions. This percentile plays key role in determining the operating characteristics of the EMAP method. Similar as logistic elastic function, we use a utility search approach to choose ρ_1 that maximizes the trade-off between power (in the congruent case) and type I error (in the incongruent case). Specifically, let ψ denote the power under the congruent case, ϕ denote the type I error under the incongruent case. We define a power and type I error trade-off utility function:

$$U(\rho_1) = \psi - w_1\phi - w_2 * (\phi - r) * I(\phi > r) \tag{12}$$

where w_1 and w_2 are pre-specified weights. The value of ρ_1 that maximizes $U(\rho_1)$ is determined using grid search, and details are provided in A.2.

Note that the percentile ρ_1, and its resulting cutoff value d are dependent on sample sizes in observed reference data n_r and test data n_t. This adds more computation complexity to the adaptive trial design, since sample sizes are determined adaptively at each interim analysis, and differ in each simulated trial. To accelerate the speed of computation after the enrollment starts, we conduct grid search beforehead with possible combinations of n_r and n_t, and provide a list of cutoff values d with different sample sizes. When real trial starts, clinicians just need to compare the sample size of real data to the list, and obtain the corresponding d value to construct the EMAP prior model. The d values implemented in simulation studies (Tables 2 and 3) are provided in Table A4.

Let $(\rho_1^{(1)}, \ldots, \rho_1^{(I)})$ denote the searching grid for ρ_1. For example, we used $\rho_1^{(1)} = 0.02$ and $\rho_1^{(I)} = 0.98$, and set grid interval of 0.02. The following steps are taken to find ρ_1 that optimizes the utility $U(\rho_1)$.

(1) Given a specific grid $\rho_1^{(i)}$, determine the values of d.
(2) Using the defined d, find the probability cutoff e that calibrates the type I error in congruence case $(\mu_r = \hat{\mu}_0)$ to the nominal value of 0.05. Compute the Power ψ in such scenario through simulation.
(3) Using the cutoff e, compute the type I error ϕ in not congruence case $(\mu_r = \hat{\mu}_0 + 0.1)$.
(4) Identify $\rho_1^{(i*)}$ that maximizes the utility function $U = \psi - w_1\phi - w_2 * (\phi - r) * I(\phi > r)$.

A.2. Results of Sensitivity Analysis

Tables A1 and A2 show the operating characteristics of the proposed AEMAP design that uses the Bayesian predictive p-value as congruence measure S with 2 interim analyses (Table A1) and 1 interim analysis (Table A2). Table A3 shows the results of sensitivity analyses with different design parameters used in the calibration procedure of the EMAP prior method, as the details described in A.1. Case 1 applies parameters ($w_1 = 1$, $w_2 = 100$, $r = 0.1$) and Case 2 applies parameters ($w_1 = 1$, $w_2 = 2$, $r = 0.15$).

Table A1. Simulation results of the AEMAP design with Bayesian predictive p-value as congruence measure and the group sequential (GS) design (without information borrowing) with 2 interim analyses. PCB denotes the percentage of claiming biosimilarity (i.e., rejecting the null hypothesis), and PET denotes the percentage of early termination.

			PCB		PET		Sample size	
Scenario	μ_r	μ_t	GS	AEMAP	GS	AEMAP	GS	AEMAP
Congruence								
1	0.413	$\mu_r - 0.1^*$	0.049	0.031	0.118	0.118	955	697
2	0.413	$\mu_r + 0.1^*$	0.053	0.040	0.114	0.098	957	704
3	0.413	$\mu_r - 0.15^*$	0.001	0.000	0.537	0.607	729	538
4	0.413	$\mu_r + 0.15^*$	0.000	0.001	0.505	0.561	744	551
5	0.413	μ_r	0.897	0.938	0.000	0.001	1020	745
6	0.392	μ_r	0.897	0.910	0.004	0.001	1020	761
7	0.434	μ_r	0.879	0.888	0.000	0.001	1020	765
Incongruence								
8	0.513	$\mu_r - 0.1^*$	0.054	0.111	0.114	0.128	956	912
9	0.313	$\mu_r + 0.1^*$	0.051	0.102	0.110	0.133	959	905
10	0.513	$\mu_r - 0.15^*$	0.001	0.022	0.497	0.510	755	703
11	0.313	$\mu_r + 0.15^*$	0.000	0.011	0.511	0.547	743	674
12	0.513	μ_r	0.870	0.835	0.000	0.001	1020	984
13	0.313	μ_r	0.935	0.855	0.000	0.000	1020	981

*PCB is type I error

Table A2. Simulation results of the AEMAP design with Bayesian predictive p-value as congruence measure and the group sequential (GS) design (without information borrowing) with 1 interim analysis. PCB denotes the percentage of claiming biosimilarity (i.e., rejecting the null hypothesis), and PET denotes the percentage of early termination.

			PCB		PET		Sample size	
Scenario	μ_r	μ_t	GS	AEMAP	GS	AEMAP	GS	AEMAP
Congruence								
1	0.413	$\mu_r - 0.1^*$	0.050	0.034	0.101	0.108	969	746
2	0.413	$\mu_r + 0.1^*$	0.055	0.039	0.108	0.134	966	740
3	0.413	$\mu_r - 0.15^*$	0.0004	0.000	0.457	0.606	787	617
4	0.413	$\mu_r + 0.15^*$	0.0004	0.000	0.454	0.556	789	632
5	0.413	μ_r	0.909	0.944	0.000	0.001	1020	778
6	0.392	μ_r	0.905	0.932	0.001	0.000	1020	790
7	0.434	μ_r	0.897	0.916	0.001	0.001	1020	797
Incongruence								
8	0.513	$\mu_r - 0.1^*$	0.051	0.103	0.106	0.141	967	933
9	0.313	$\mu_r + 0.1^*$	0.051	0.113	0.096	0.115	972	942
10	0.513	$\mu_r - 0.15^*$	0.000	0.014	0.461	0.514	785	743
11	0.313	$\mu_r + 0.15^*$	0.000	0.016	0.451	0.488	791	752
12	0.513	μ_r	0.878	0.867	0.000	0.000	1020	1003
13	0.313	μ_r	0.933	0.897	0.000	0.000	1020	1001

*PCB is type I error

Table A3. Results of sensitivity analyses with different design parameters (Case 1 ($w_1 = 1$, $w_2 = 100$, $r = 0.1$) and Case 2 ($w_1 = 1$, $w_2 = 2$, $r = 0.15$)) used in the calibration procedure of the EMAP prior method.PCB denotes the percentage of claiming biosimilarity, and PET denotes the percentage of early termination.

			PCB		PET		Sample size	
Scenario	μ_r	μ_t	Case 1	Case 2	Case 1	Case 2	Case 1	Case 2
Congruence								
1	0.413	$\mu_r - 0.1^*$	0.028	0.036	0.116	0.105	706	698
2	0.413	$\mu_r + 0.1^*$	0.035	0.038	0.115	0.110	707	697
3	0.413	$\mu_r - 0.15^*$	0.001	0.001	0.625	0.622	533	525
4	0.413	$\mu_r + 0.15^*$	0.000	0.000	0.580	0.582	553	543
5	0.413	μ_r	0.933	0.956	0.001	0.001	754	739
6	0.392	μ_r	0.908	0.925	0.001	0.001	768	751
7	0.434	μ_r	0.885	0.908	0.002	0.002	775	754
Incongruence								
8	0.513	$\mu_r - 0.1^*$	0.100	0.132	0.133	0.133	913	896
9	0.313	$\mu_r + 0.1^*$	0.100	0.125	0.115	0.115	931	912
10	0.513	$\mu_r - 0.15^*$	0.012	0.023	0.531	0.521	699	685
11	0.313	$\mu_r + 0.15^*$	0.014	0.024	0.527	0.510	706	694
12	0.513	μ_r	0.837	0.817	0.003	0.004	990	972
13	0.313	μ_r	0.887	0.863	0.002	0.003	994	974

*PCB is type I error

Table A4. The value of d in the step elastic function used in simulation studies (Tables 2 and 3), obtained using the proposed calibration procedure.

n_r	n_t	d
170	170	0.4384
170	340	0.3522
340	340	0.4701
210	510	0.3929
340	510	0.4444
510	510	0.4829
255	255	0.4697
255	510	0.3897